CHE 230

1. **Course number and name**
   CHE 230. Molecular Systems in Chemical Engineering

2. **Credits, contact hours and categorization**
   Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. **Instructor's or course coordinator's name**
   Roshan Nemade, Ezinne Achinivu

4. **Textbook, title, author, publisher, year**
   Materials Science and Engineering: An Introduction, Willam D. Callister Jr and David G. Rethwisch, Wiley

5. **Specific course information**
   a. Brief description of the content of the course (catalog description)
   Introduction to fundamental concepts in molecular engineering and materials chemistry; properties of molecular systems and applications of macromolecules, bio-macromolecules and nanomaterials in energy, medicine, environment and technology.
   b. Prerequisites or co-requisites
   CHEM 232.
   c. Indicate whether a required, elective, or selected elective.
   Required.

6. **Specific goals for the course**
   a. Specific outcomes of instruction.
   The main objective of the course is to familiarize the students with the fundamental concepts of Materials Science and Engineering which will be used as background knowledge for the understanding of specialized problems in the field of Chemical Engineering. At the end, students will be able to:Understand the fundamental and essential concepts used in polymeric materials for Chemical Engineering applications.
      1. Understand the fundamental and essential concepts used in materials chemistry and molecular systems
      2. Classify the molecular systems, nanomaterials, and materials systems.
      3. Know the basic methods to synthesize and characterize materials systems
      4. Understand the basic properties that characterize the behavior of molecular systems
      5. Select appropriate type of material for specific application
      6. Offer different approaches to modify structure/microstructure to get desired properties

   One semester long project to reflect the three components of molecular engineering related to Polymer Design, Processing and Application. Students will be working with groups and are expected to prepare a case study. Each group will be assigned one molecular system application to perform a critical review to explore products material properties and performance requirements from a molecular system lens.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.			X	
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X		
(3)	an ability to communicate effectively with a range of audiences.		X		
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.		X		
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.		X		

7. Brief list of topics to be covered

- Introduction to Molecular Systems
- Molecular Structure & Defects
- Diffusion & Material Properties from molecular structure
- Molecular Interactions: Influence on Macroscopic Properties
- Engineering Functional Materials
- Introduction to Macromolecules
- Nanoscale Systems
- Sustainability, Circular Materials & commercializing products with molecular engineering